starRoboHarness

An Open Execution Harness for Reasoning-Driven Robotics

Task & World State

- **Task**
- **Observations**
- **Robot State**
- **Action Constraints**

In-Context Learning

Ongoing: efficiency & reuse

- **Demonstrations** — Task examples
- **References** — Visual and task context
- **History / Feedback** — Prior traces and outcomes

Context

starRoboHarness Core

High-Level Vision–Language Reasoner

GPT · Claude · Gemini · Qwen · open models | Optional policy-only

Execution Profiles & Authority Boundaries

Adaptive routing: ongoing

VLM Direct Control	Hybrid Policy Supervision	Agentic Tool Use	VLA / WAM Policy
GPT-6	GPT-6 + starVLA	SAM · GraspGen	starVLA · π0.5 · OpenWAM
Direct actions	Review / correction	Tool adapters	

Action proposals

Evidence Gate — Identity · freshness · action contract | Accept · correct · reject

Accepted actions

Unified Executor — Single controller → **Robot / Environment Adapters** → **ACKs** → **Execution evidence**

Observability, Evaluation & Audit

- **Execution Monitoring** — Traces · metrics · token / time usage
- **Native Evaluation** — RoboDojo · BEHAVIOR · SimplerEnv
- **Contract & Risk Checks** — Identity · freshness · action bounds
- **Execution Trace & Native Receipt** — Decisions · ACKs · usage · outcome

Review outcomes

Modular Interfaces & Extension Points

- **Reasoner Interfaces** — Model-agnostic
- **Tool Adapters** — VLA · WAM · classical control
- **Benchmark Adapters** — Native actions & evaluation
- **Robot Adapters** — Physical validation: future

Curated feedback

Human-in-the-Loop Iterative Improvement

Curated context · new adapters · audited recovery rules

Community · **Academia** · **Researchers** · **Industry**

Solid: current interfaces

Dashed: ongoing / future